

Sreerag M

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🔗 <https://github.com/Sree-ragM>

👤 PROFILE

Experience in computational chemistry, developing a specialized Python tool to predict the hydrogen bond propensity of drug molecules. Proficient in programming, a collaborative team player, and dedicated to continuous learning. Achieved second authorship by publishing a research article using my skills in Python programming.

🎓 EDUCATION

2024 – 2026 Kollam	Master of Science in Bioinformatics Amrita School of Biotechnology
2020 – 2023 Kollam	Bachelor of Science in Biotechnology Amrita School of Biotechnology
2018 – 2020 Thrissur	Higher Secondary Examination MR Raman Memorial Higher Secondary School
2017 – 2018 Thrissur	All India Secondary School Examination Guruvayoor Devaswom English Medium School

📖 PUBLICATIONS

2024	Quantifying the hydrogen-bond Propensity of Drugs and Relationship to Lipinski's Rule of Five , <i>New Journal of Chemistry</i> https://doi.org/10.1039/D3NJ05476D 🔗
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🧠 SKILLS

Python Programming	Pipeline Development
Web Development	Cloud Management and Dockerization
Version Control	Database Management

📁 PROJECTS

01/2026 – Present	Proteo-transcriptomics and machine learning approaches for the identification of species-specific antigenic epitopes from Indian Cobra venom The work focuses on developing a computational pipeline for generating predicted proteomes from venom transcriptomic data, applying machine learning-based protein structure prediction, and identifying antigenic epitopes.
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07/2025 – 12/2025

Petrinet

This project focuses on automatically detecting and counting bacterial colonies in images of Petri dishes. It uses image processing techniques and machine learning models to identify colonies from masks and estimate their number accurately. The system helps reduce manual effort and improves consistency in colony counting. It also handles challenges like overlapping colonies using advanced methods.

01/2023 – 06/2023

Cerebellar Input-Output Neuron Modelling

Limited understanding of the olivo-cerebellar system's role in motor learning and neurodegenerative diseases like Ataxia has challenges in advancing research and treatment. To address this, we developed a model integrating experimental data, neuron modeling, and simulations to produce testable predictions. This model yielded new insights into the olivo-cerebellar system's function, enhancing our knowledge of motor learning. Dr Shyam Diwakar and Amrita School of Biotechnology approved this project.

INTERNSHIPS

2023 – 2024

Thiruvananthapuram

HBP-PREDICTOR, CSIR-NIIST

Developed a tool in Python to identify water-binding motifs and hydrogen-bond propensity. Used RDKit, NumPy, Math, and Regular Expression to develop the tool.

TECHNICAL EXPERIENCE

ASBT Kalotsavam Event Management Web Application, Amrita School of Biotechnology

ASBT Kalotsavam web application is a paperless event management platform with student registration, admin-controlled winner allotment, and real-time leaderboard for Amrita School of Biotechnology's annual arts fest.

Lipinski Lens, <https://lipinski-lens.up.railway.app/>

Lipinski Lens is an intuitive web-based tool that evaluates the drug-likeness of chemical compounds using Lipinski's Rule of Five.

Biolympics Event Management Web Application, Amrita School of Biotechnology

The Biolympics web application is a paperless event management platform with student registration, admin-controlled winner allotment, and a real-time leaderboard for Amrita School of Biotechnology's annual sports meet.

AREA OF INTEREST

Bio Informatics, Machine Learning, Deep Learning, Proteomics, Venomics

COMMUNITY OUTREACH PROGRAMS

2023 – 2023

Analyzing Healthcare, Waste Management, and Women Empowerment Nexus in Community

Actively participated in a sustainable development project at Karumattoor, Tamil Nadu, India collaborated with Amrita Center for Wireless Networks & Applications and UNESCO Chair and received A grade with UNESCO Chair Certificate.

WORKSHOPS

2025 – 2025	E-workshop on Basic Bioinformatics Attended 3-day Hands-on workshop on basic bioinformatics tools and topics.
2024 – 2024	Leveraging Machine Learning to Predict Drug Targets Attended a one-day workshop on the topic of leveraging machine learning to predict drug targets
2024 – 2024	PyMol for Beginners-Guide to Create Stunning Molecular Graphics Attended one-day Hands-on workshop on the topic PyMol for Beginners Guide to Create Stunning Molecular Graphics.
2023 – 2023	Fundamentals of Molecular Biology & Cancer Tools & Techniques Attended a 3-day online workshop on fundamentals of molecular biology and cancer tools and techniques
2023 – 2023	International Workshop on Genome Informatics 2023 Attended a 1-month Workshop on Genome Informatics.

CERTIFICATES

AI for Medicine

Focusing on machine learning applications in healthcare, including diagnostic imaging (X-ray, MRI), survival analysis using tree-based models, treatment effect estimation from randomized controlled trials, and NLP-based extraction of clinical information from unstructured data.

EXTRA-CURRICULAR

2026	First Prize in Basketball Tournament (2025 & 2026) and Best Player BIOLYMPICS (Amrita School of Biotechnology)
2026	First prize in Volleyball Tournament (2025 & 2026) and Best Player BIOLYMPICS (Amrita School of Biotechnology)
2025	Inter-Amrita Basketball Player Participated in the Inter-Amrita Basketball Tournament 2025 held at Coimbatore, serving as the Captain of the B Team representing Amritapuri Campus.
2024	Team Lead Serving as a Developer in the Society for Interdisciplinary Biologists, Amrita School of Biotechnology.

REFERENCES

Dr. Nidheesh M.,

Principal, Amrita School of Physical Sciences | Associate Professor, Amrita School of Biotechnology, Amritapuri
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Dr. Muralidharan V., *Assistant Professor, Amrita School of Biotechnology, Amritapuri*
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